

SANZIDA AKTER

Ph.D. Candidate in Electrical Engineering, University of Maryland, Baltimore County, USA

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Research Profile

Fourth-year Ph.D. candidate in Electrical Engineering with research expertise in nonlinear optics, computational physics, and topological photonics. My work focuses on using analytical and numerical approaches as tools to study the nonlinear dynamics and stability of complex optical systems, with particular emphasis on microresonator-based frequency combs, nonlinear wave dynamics, and topological edge-state solitons. I am interested in translating my computational and photonics expertise into application-driven technologies and in extending these approaches to interdisciplinary problems where robust modeling, precision measurement, and integrated photonic systems can enable real-world impact.

Research Interests

Nonlinear Optics and Photonics Computational Photonics Microresonator Frequency Combs

Topological Photonics Numerical Modeling of Nonlinear PDEs Quantum Photonic Technologies

Education

Ph.D. in Electrical Engineering **Expected August 2027**
University of Maryland, Baltimore County, USA

- Advisor: Professor Curtis R. Menyuk

M.S. in Electrical Engineering **December, 2025**
University of Maryland, Baltimore County, USA *GPA: 3.7/4.00*

M.S. in Electrical and Electronic Engineering **January 2021–August, 2022**
University of Dhaka, Dhaka, Bangladesh *GPA: 3.69/4.00*

B.S. in Electrical and Electronic Engineering **January, 2016–December, 2020**
University of Dhaka, Dhaka, Bangladesh *GPA: 3.66/4.00*

Certifications and Specialized Training

AIM Photonics Certified Training in Photonic Integrated Circuits (PIC) Fall 2025

- Training in integrated photonic circuit concepts, design, fabrication, and device-level implementation.

Research Experience

Graduate Research Assistant **August, 2022–Present**
Computational Photonics Laboratory *Halethorpe, MD*

- Conduct theoretical and computational research in nonlinear optics and photonics, focusing on optical pattern formation, dissipative Kerr solitons, microresonator frequency combs, and nonlinear topological photonic systems.
- Develop reduced-order and scalable computational methods for nonlinear optical systems, including approaches for long-timescale soliton dynamics, stochastic simulations, and large-scale stability analysis.
- Investigate nonlinear edge-localized states in topological microring lattices, combining numerical mode reduction, spectral analysis, perturbation theory, Floquet analysis, and Lyapunov diagnostics to characterize stationary, periodic, and chaotic dynamics.
- Collaborate across interdisciplinary research teams on analytical and computational studies of nonlinear photonic systems, contributing numerical modeling, and physical interpretation.

Selected Current Research Projects

Noise-Driven Dynamics and Experimental Validation of Optical Soliton Patterns **May, 2025–Present**
Lead Graduate Researcher

- **Objective:** Investigate how noise influences the long-term stability and evolution of dissipative soliton structures in nonlinear optical microresonators, and connect numerical predictions with experimentally observed soliton behavior.
- **Approach:** Extend the synergetic simulation framework to model stochastic soliton dynamics over experimentally relevant timescales, using noise-driven simulations and statistical stability analysis to compare predicted soliton structures and transitions with experimental observations.

Stability and Noise Dynamics of Topological Frequency Combs

November, 2025–Present

Lead Graduate Researcher

- **Objective:** Develop a broader understanding of the formation, stability, and robustness of edge-localized nonlinear states and nested solitons in topological microring frequency-comb systems.
- **Approach:** Model nonlinear dynamics in anomalous-quantum-Hall-effect microring lattices using coupled Lugiato–Lefever equations, numerical mode reduction, and spectral stability analysis.

Efficiency Enhancement of Soliton Microcombs through Mode Coupling

January, 2026–Present

Research Mentor and Project Contributor

- **Objective:** Develop an analytical understanding of how avoided mode crossings and intermodal coupling modify dissipative Kerr soliton dynamics and enhance pump-to-comb conversion efficiency.
- **Approach:** Investigate the dependence of soliton efficiency, accessible detuning, and stability on mode splitting and coupling strength using coupled-mode theory, analytical modeling, and nonlinear Lugiato–Lefever simulations.

Publications

Peer-Reviewed Journal Articles

1. **S. Akter**, R. Anguluri, and C. R. Menyuk, “Model-Order Reduction in Microresonators Using Proper Orthogonal Decomposition.” *Journal of the Optical Society of America B*, vol. 43, no. 8, pp. 1854–1860, 2026.
2. A. Leshem, **S. Akter**, L. Courtright, P. Shandilya, C. R. Menyuk, and O. Gat, “Dynamics of Interacting Cavity Solitons.” *Physical Review E*, vol. 113, no. 1, 014207, 2026.

Manuscripts in preparation

1. **S. Akter**, P. Shandilya, L. Courtright, A. Leshem, G. D’Aguanno, O. Gat, and C. R. Menyuk, “Fast Synergetic Simulation to Study Slow Evolution of Soliton Patterns in Optical Resonators.”, 2026.
2. **S. Akter**, L. Xu, M. J. Mehrabad, M. Hafezi, and C. R. Menyuk, “Stability Analysis of Nested Solitons in Topological Microresonator Lattices.”, 2026.

Conference Papers

1. **S. Akter**, L. Xu, M. J. Mehrabad, M. Hafezi, and C. R. Menyuk, “Chaos Diagnostics in Topological Super-Ring Kerr Microcombs via Lyapunov Exponents.” In *CLEO 2026*, Optica Publishing Group, 2026.
2. L. Courtright, A. Sukhinin, P. Shandilya, **S. Akter**, and C. R. Menyuk, “Bistability of Platonic Solutions in Normal Dispersion Photonic Crystal Resonators.” In *Laser Resonators, Microresonators, and Beam Control XXVIII*, Proceedings of SPIE, 2026.
3. **S. Akter**, P. Shandilya, L. Courtright, O. Gat, and C. R. Menyuk, “Efficient Synergetic Simulation Method for Studying Soliton Molecule Formation on Laboratory Timescales.” In *Frontiers in Optics + Laser Science 2025 (FiO+ LS)*, Optica Publishing Group, 2025.
4. **S. Akter**, P. Shandilya, L. Courtright, G. D’Aguanno, R. Anguluri, O. Gat, and C. R. Menyuk, “Fast Synergetic Simulation of Soliton Molecules in Microresonators in the Presence of Noise.” In *Laser Resonators, Microresonators, and Beam Control XXVII*, Proceedings of SPIE, 2025.
5. **S. Akter**, P. Shandilya, L. Courtright, G. D’Aguanno, A. Leshem, O. Gat, and C. R. Menyuk, “An Efficient Computational Algorithm for Modeling Slow Soliton Interactions in Microresonators.” In *Frontiers in Optics + Laser Science 2024 (FiO+ LS)*, Optica Publishing Group, 2024.

Awards

Best Graduate Student Poster Presentation Award, University of Maryland, Baltimore County — awarded three consecutive years 2024–2026

Teaching Experience

University of Maryland, Baltimore County (UMBC)

- Teaching Assistant & Laboratory Instructor, CMPE 306: Introductory Circuit Theory (Fall 2022–Spring 2023)
- Laboratory Instructor, CMPE 314: Electromagnetic Theory (Fall 2024)
- Teaching Assistant, ENEE 605: Applied Linear Algebra (Spring 2025)

Mentoring Experience

- Mentor a Ph.D. student conducting theoretical and computational research on improving dissipative Kerr soliton conversion efficiency through avoided mode crossings and intermodal coupling in optical microresonators (Fall 2025–Present)

Leadership and Professional Service

- Selected Session Presider, Conference on Lasers and Electro-Optics (CLEO), 2026.
- Reviewer, *IEEE Journal of Quantum Electronics*, Fall 2025–Present.
- Reviewer, *Optics Express*, Summer 2026–Present.
- Organizing Secretary, SPIE Student Chapter at University of Maryland, Baltimore County (Fall 2025–Present)

Technical Skills

Scientific Tools	MATLAB, Python, COMSOL Multiphysics
Scientific Computing	Numerical solution of nonlinear PDEs, eigenvalue and spectral analysis, reduced-order modeling, Monte Carlo simulation, stochastic modeling, stability analysis
Nonlinear Photonics	Lugiato–Lefever equation modeling, coupled-mode theory, nonlinear cavity dynamics
Topological Photonics	Coupled microring lattice modeling, topological edge-state dynamics
Numerical Methods	Split-step Fourier methods, Runge–Kutta methods, Jacobian-based linearization, proper orthogonal decomposition, mode-reduction methods

Professional Memberships

- Optica Student Chapter
- APS + DLS Student Member
- IEEE Student Member

References

Dr. Curtis R. Menyuk

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